

Domain Frequency Analysis for Suspicious Domain Detection

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Project Overview

This case study demonstrates a frequency-based approach for identifying uncommon domain names that may warrant additional security investigation. Statistical analysis of domain labels is widely used during threat hunting to prioritize domains for deeper inspection, particularly when investigating Domain Generation Algorithm (DGA) activity or anomalous DNS behavior. Rather than attempting to classify domains as malicious, this project establishes a repeatable methodology for identifying lexically uncommon domain names using character frequency scoring.

Objective

The objective of this project was to analyze a ranked domain dataset, extract the primary domain labels, apply a character frequency scoring algorithm, and identify domains whose statistical characteristics differ significantly from commonly observed domains. The resulting list represents candidate domains that may require further investigation using threat intelligence, DNS telemetry, or behavioral analysis.

Methodology

A ranked domain dataset was processed using Python and Jupyter Notebook. The `tlextract` library was used to isolate the primary domain label while removing subdomains and top-level domains. Each unique label was evaluated using a frequency-based scoring algorithm derived from character occurrence statistics. Duplicate labels were removed before calculating the final count of domains below the selected threshold of 3.5. This workflow demonstrates a practical preprocessing technique used during security analytics and threat hunting.

Tools and Technologies

Python, Jupyter Notebook, Pandas, and `tlextract` were used to perform data parsing, preprocessing, statistical analysis, and result generation.

Results

The analysis successfully processed the dataset and identified 32,898 unique domain labels that scored below the selected threshold. These domains represent statistically uncommon labels that can be prioritized for additional investigation. Although low-frequency scores alone do not indicate malicious intent, they provide an effective filtering mechanism for reducing large datasets into manageable investigation candidates.

Security Relevance

Modern security operations centers analyze millions of DNS events every day. Frequency-based analysis can assist analysts by highlighting domains whose lexical characteristics differ from legitimate domains. Combined with DNS logs, passive DNS, threat intelligence, and behavioral indicators, this technique contributes to more efficient threat hunting workflows and early detection of suspicious activity.

Limitations

This project evaluates lexical characteristics only and does not determine whether a domain is benign or malicious. Additional evidence such as DNS resolution history, WHOIS information, passive DNS records, or threat intelligence feeds would be required before making a security determination. Furthermore, the original dataset is not distributed with this repository due to course licensing and data-sharing restrictions.

Conclusion

This project demonstrates how statistical domain frequency analysis can support cybersecurity investigations by identifying uncommon domain labels for further analysis. The implementation highlights practical data preprocessing, feature extraction, and analytical techniques that can be incorporated into enterprise threat hunting workflows. It also establishes a foundation for more advanced domain-based detection methods, including machine learning and DGA detection.